Cantera vs NN | Test IC 76 | NRMSE (util.py)

T0 = 1523.6 K P0 = 3.626 atm phi0 = 1.349 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	7.28e-03	0.157	0.733	1.66	0.164	0.967	0.0661	3.08	10.3	0	Complete
2 P-only DeepONet10	**10k**	0.0606	0.124	0.537	0.631	0.585	0.637	0.075	1.43	3.56	0	Complete
2 P-only DeepONet10	**long**	5.46e-03	0.0458	0.124	0.194	0.173	0.117	0.0176	0.174	0.215	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.0447	0.119	0.469	0.535	0.487	0.537	0.062	1.43	4.32	0	Complete
3 ANN10 + P-only DeepONet10	**long**	7.37e-03	0.0772	0.224	0.293	0.27	0.199	0.0288	0.178	0.333	0	Complete
4 T/P DeepONet10	**10k**	0.101	0.138	1.02	0.564	0.519	1.13	0.102	4.47	2.83	0	Complete
4 T/P DeepONet10	**long**	0.502	0.507	6.45	0.0877	0.396	3.32	0.259	1.13	0.876	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.0593	0.113	0.713	0.206	0.165	0.838	0.067	0.92	1.56	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.224	0.0196	1.71	0.778	0.613	3.06	0.222	5.46	5.78	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.0794	0.0678	0.632	0.485	0.728	0.848	0.0779	2.08	2.75	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.215	0.0118	1.85	0.378	0.196	2.51	0.195	2.63	2.06	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.0672	0.0325	0.52	0.132	0.106	0.894	0.0693	0.916	1.18	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.167	0.0121	1.31	0.565	0.383	2.23	0.164	1.54	1.7	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.